25900275

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular Examination October 2025

Third Semester

Value Addition Course (VAC) - MG3VACBTG200, MG3VACBBT201 -

ENVIRONMENTAL BIOTECHNOLOGY AND HUMAN RIGHTS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-One word or sentence-Answer any 10 questions

Each question carries 2 marks

1. Explain the importance of the Water Act in controlling industrial pollution. [U] / [CO1]
   Give any 2 points about water act- 2 marks.
2. Name any two main sources of wastewater. [K] / [CO2] –
   Mention any 2 sources of waste water- 2 marks
3. Give two examples of recyclable solid wastes. [K] / [CO3]
   Any 2 examples of recyclable solid wastes-2 marks
4. Which gas causes foul odor in wastewater? [K] / [CO2]- Mention any gas that causes foul odour in water- 2 marks
5. What is the main role of a sedimentation tank? [U] / [CO3]
   Mention 2 roles of sedimentation tank- 2 marks
6. Name any four sources of solid waste. [K] / [CO3]
   4 sources – 2 marks
7. Give two examples of polycyclic aromatic hydrocarbons (PAHs). [U] / [CO2]
   Any 2 examples – 2 marks.
8. Name one indicator organism of pollution. [K] / [CO2]
   Any one indicator organism -2 marks
9. The process of using microbes to treat wastewater? [U] / [CO3]
   Biological treatment or bioremediation or biodegradation- 2 marks
10. Name two microorganisms involved in composting. [U] / [CO2]

Any one microbe involved in composting-2 marks

11. Which treatment step follows primary treatment? [U] / [CO3]

    Secondary treatment- 2 marks

12. Which is used as an alternative to petrol and diesel for transport vehicles? [U] / [CO1]

    Any alternative source to petrol and diesel- 2 marks

[2x10 = 20]

Part B-Short Essay-Answer any 5 questions

Each question carries 4 marks

13. The producers considered the most important biotic component in an ecosystem. Justify the statement.[U] / [CO1]

    Producer – Definition-1 mark, food chain- 2 marks, explanation- 1mark.

14. Describe the 10% law of energy transfer with an example. [K] / [CO1]

    Define 10% law -1 mark, explain with example -3 marks

15. Describe why a climax community has high stability and low change. [K] / [CO1]

    Climax community – 1mark, ecological succession –explanation- 3 marks

16. Define carbon cycle and its specialties. [U] / [CO1]

    Carbon cycle- Definition -1 mark, explanation -3 marks

17. Examine the impact of human activities on the balance of an ecosystem [K] / [CO1]

    4 activities and its impacts- 4 marks

18. Explain the importance of each trophic level in a food chain. [K] / [CO1]

    Food chain-definition 1- components-explain -3 marks.

19. Explain the main function of State Human Right Commission [U] / [CO4]

    Any 4 functions -4 marks.

[4x5 = 20]

Part C-Essay type questions-Answer any 1 questions

Each question carries 10 marks

20. Discuss the role of the UN Human Rights Council (UNHRC) in monitoring and addressing human rights violation? [U] / [CO4]- UNHRC explanation -3 marks, Role-explain- 7 marks

21. Explain the significance of French Declaration of the Rights of Man and Citizen? [U] / [CO4]

    French declaration -3 marks, significance -7 marks

[10x1 = 10]